Template Document

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Show that there exists no nontrivial unramified extensions of .

If $K/$ is a nontrivial number field, then $|K| > 1$. But then K has a prime factor so that some prime ramifies in K.

Complete the following:

1. How does one prove a cotheorem?
2. Compute $\int \cos x \; dx$.
3. How does one square $abcd$?

1. Use rollaries.
2. We have

$$\int \cos x \; dx = \sin x + C \tag{1}$$

   We can check (1):

$$\frac{d}{dx} \left(\sin x + C \right) = \cos x$$

3. This is routine.

Prove that $\sqrt{2}$ is irrational.

Assume that $\sqrt{2} = \dfrac{a}{b}$, where $a, b \in$. Without loss of generality, we may assume $\gcd(a, b) = 1$. Then we have

$$\begin{aligned} \sqrt{2} &= \frac{a}{b} \\ \sqrt{2}^2 &= \left(\frac{a}{b} \right)^2 \qquad (2) \\ 2 &= \frac{a^2}{b^2} \\ a^2 &= 2b^2 \qquad (3) \end{aligned}$$

But then from (3), we know that a^2 is even so that a is even. But then we must have

$$2a^2 = b^2$$

so that b^2 is even, implying b is even. But then $\gcd(a, b) \geq 2$, a contradiction.